

AC7922A Datasheet

Zhuhai Jieli Technology Co.,LTD

Version 1.2

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Revision History

Date	Revision	Author	Description
2024.05.30	V1.0	zh-jieli	Initial Release
2024.09.30	V1.1	zh-jieli	Add Power Domain Information
2024.12.07	V1.2	zh-jieli	Modify Audio DAC Characteristics, IO Characteristics and Operating Temperature

Table of Contents

Revision History	1
Table of Contents	2
AC7922A Features	3
1 Block Diagram	5
2 Pin Definition	6
2.1 Pin Assignment	6
2.2 Pin Description	7
3 Electrical Characteristics	12
3.1 Absolute Maximum Ratings	12
3.2 ESD Ratings	12
3.3 PMU Characteristics	12
3.4 IO Characteristics	13
3.5 Audio DAC Characteristics	16
3.6 Audio ADC Characteristics	17
3.7 BT Characteristics	18
3.8 WiFi Characteristics	18
4 Package Information	20
4.1 QFN96_10*10mm	20
5 IC Marking Information	21
6 Solder-Reflow Condition	22

AC7922A Features

SYSTEM

- Dual Core 32bit DSP 320MHz
- With IEEE754 Single precision FPU
- Support jieli TEE
- Support FFT / MATRIX / MATH
- 2 x I-cache and D-cache
- On-chip SRAM 352kbyte
- Support SDTAP / EMU / ETM
- Support MMU
- Support MPU
- Built-in SDRAM/DDR (Maximum 64Mbyte)
- SPI FLASH Controller (Maximum 64Mbyte)
- 24MHz crystal oscillator
- 32KHz RTC crystal oscillator
- Internal RC oscillator, PLL

Video Input

- Internal Image Signal Processor
- Support DVP, BT656, SPI interface
- Support 1 lane MIPI-CSI interface
- Support RAW, YUV422 formats
- Support video resize an time mark
- 2 x JPEG codec

Video output

- Support display color enhancement
- Support DPI, DBI, BT656 interface
- Support 4 lane MIPI-DSI interface
- Support RGB, YUV formats

Graphics

- Internal 2D DMA
- Internal 2.5D GPU
- Support vector graphics rendering
- Support image resize, rotation, projection
- Support multiple blending mode
- Support ARGB, RGB, YUV, Lx, Ax formats

DSP Audio Processing

- SBC/AAC/LDAC/LHDC/LC3/CVSD/mSBC codec
- mSBC voice codec supported for BT phone
- PLC for voice processing
- Single/Multi MIC ENC
- Multi-band DRC
- Multi-band EQ
- Support spatial sound

Audio

- 2 x 16bit DAC
 - ❖ SNR 103dB
 - ❖ Noise 6.4uVrms
 - ❖ Supports differential mode
 - ❖ Sampling rate 8~96kHz
- 2 x 16bit ADC
 - ❖ SNR 95dB
 - ❖ Sampling rate 8~48kHz
- I2S/PDM AUDIO Master/Slave interface

Bluetooth

- Dual-mode BT5.4 with LE Audio (QDID 223418)
- Support AoA/AoD
- Support LE audio BIS/CIS
- Support long range BLE
- Maximum transmitting power 19 dBm
- Receiver sensitivity
 - ❖ -95.5 dBm @BR
 - ❖ -96 dBm @EDR $\pi/4$ DQPSK
 - ❖ -88 dBm @EDR 8DPSK

IEEE 802.11b/g/n

- 1T1R in 2.4 GHz band
- 20 MHz and 40 MHz bandwidth
- Data rate up to 150 Mbps
- Security: WFA/WPA3 personal, WPS2.0, WAPI
- QoS: WFA WMM, WMM PS
- Support STBC, A-MPDU, A-MSDU, BLK-ACK
- Support Station, SoftAP, Station+SoftAP, Promiscuous mode
- Maximum transmitting power
 - ❖ 19 dBm @1Mbps, DSSS
 - ❖ 17 dBm @HT20, MCS0
 - ❖ 13 dBm @HT20, MCS7
- Receiver sensitivity
 - ❖ -97 dBm @1Mbps, DSSS
 - ❖ -93 dBm @HT20, MCS0
 - ❖ -74 dBm @HT20, MCS7

Peripherals

- 1 x High speed USB
- 2 x SD host controller
- 6 x Multi-function 32bit timer
- 5 x UART interface
- 3 x I²C Master/Slave interface
- 3 x SPI Master/Slave interface
- 1 x QDEC
- 1 x CAN Controller
- 1 x Ethernet Mac
- 8 x MCPWM
- 1 x PAP Interface
- 3 x Light strip Controller
- 1 x 10bit ADC(6 Channel)
- 49 x GPIO Support function remapping
- Built-in RTC with alarm, wakeup

PMU

- 1 x Buck DC-DC converter
- 2 x IO power domain
- WIFI PA support external power supply
- RTCVDD33 support external power supply
- Support temperature sensor
- VBAT range 2.7V to 5.5V
- IOVDD range 2.7V to 3.6V

Packages

- QFN96(10mm*10mm)

Temperature

- Operating temperature
TC = -20℃ to +85℃(standard range)
- Storage temperature -65℃ to +150℃

Applications

- IPC
- Driving recorder
- WIFI Appliances

1 Block Diagram

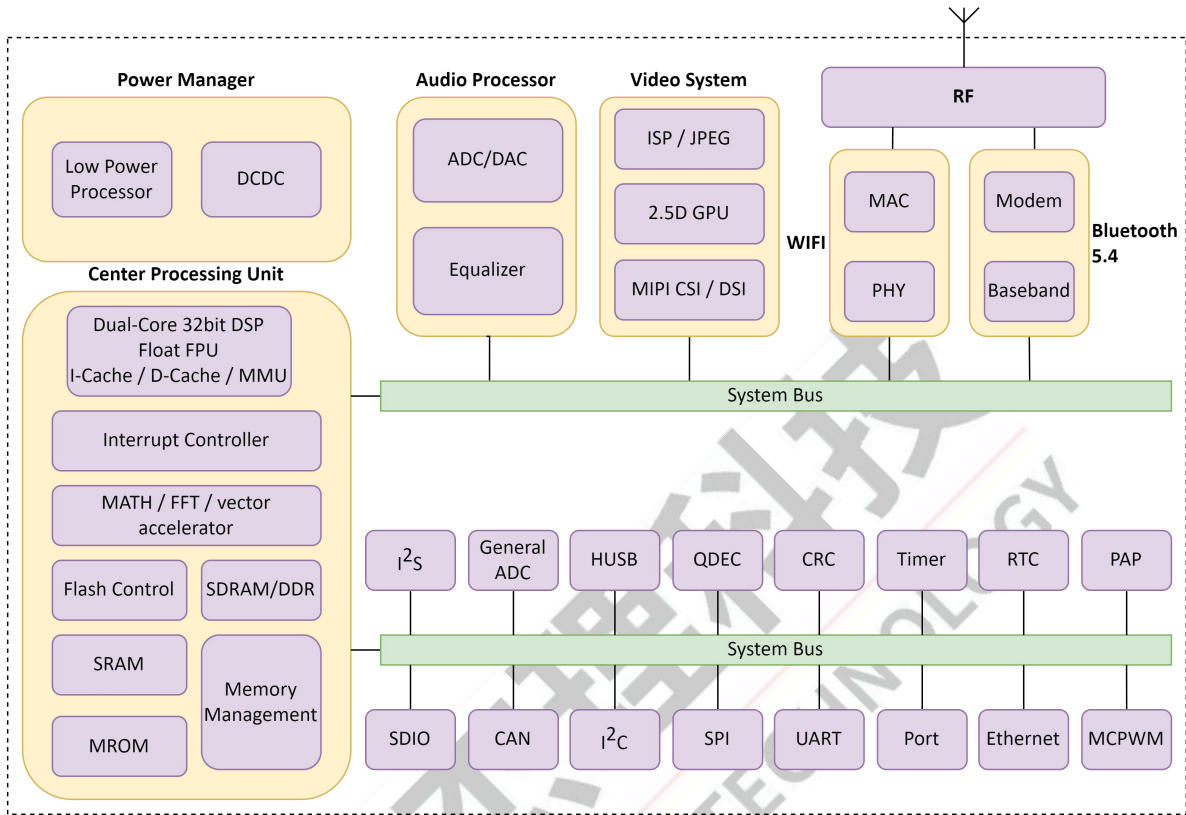


Figure 1-1 AC7922A Block Diagram

2 Pin Definition

2.1 Pin Assignment

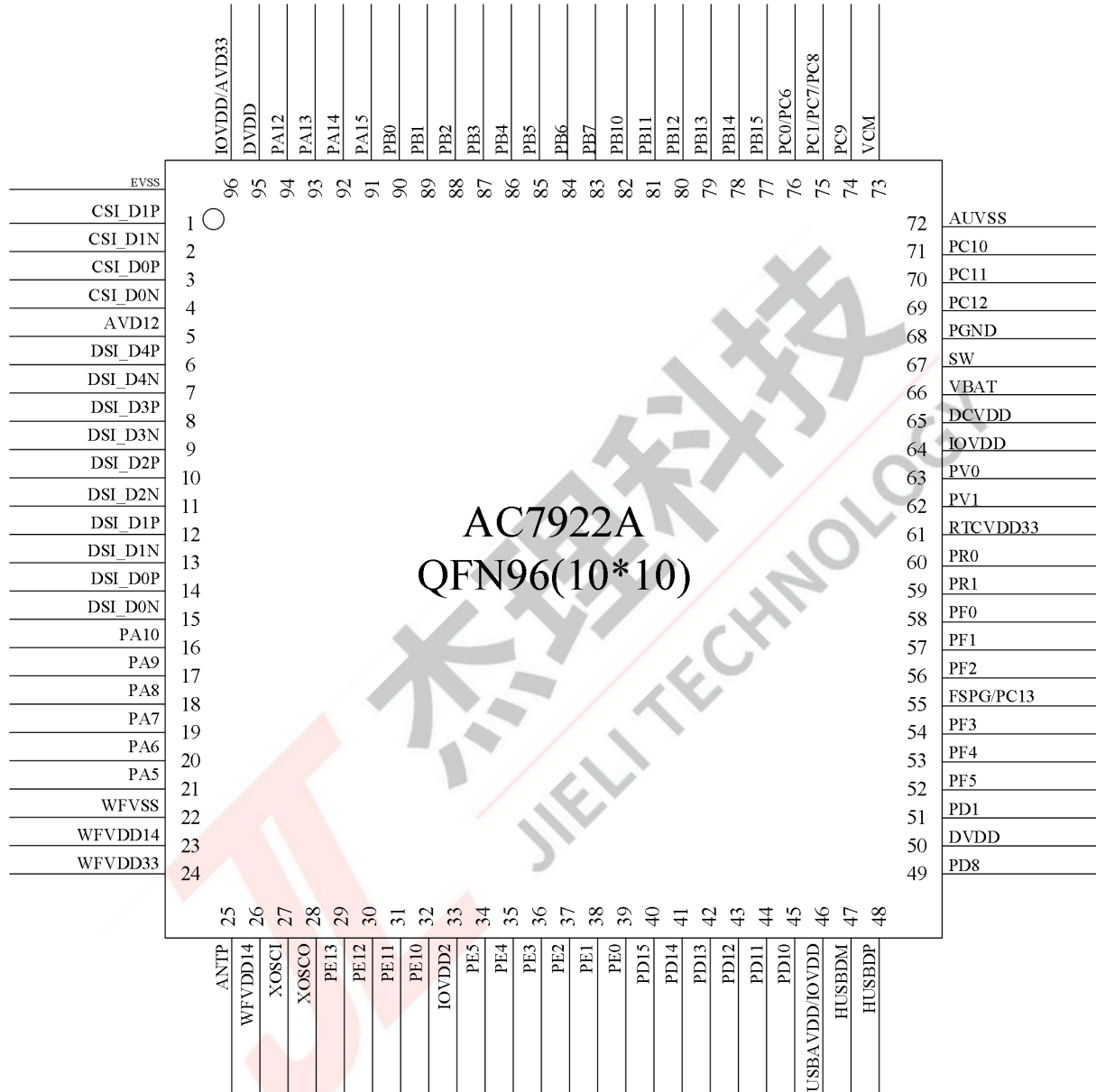


Figure 2-1 AC7922A Pin Assignment

2.2 Pin Description

Table 2-2-1 AC7922A Pin Description

Pin No.	Name	Type	IO Initial State	Description
1	CSI_D1P	I	--	MIPI CSI D1P
2	CSI_D1N	I	--	MIPI CSI D1N
3	CSI_D0P	I	--	MIPI CSI D0P
4	CSI_D0N	I	--	MIPI CSI D0N
5	AVD12	P	--	Analog 1.2V Power
6	DSI_D4P	I/O	--	MIPI DSI D4P
7	DSI_D4N	I/O	--	MIPI DSI D4N
8	DSI_D3P	I/O	--	MIPI DSI D3P
9	DSI_D3N	I/O	--	MIPI DSI D3N
10	DSI_D2P	I/O	--	MIPI DSI D2P
11	DSI_D2N	I/O	--	MIPI DSI D2N
12	DSI_D1P	I/O	--	MIPI DSI D1P
13	DSI_D1N	I/O	--	MIPI DSI D1N
14	DSI_D0P	I/O	--	MIPI DSI D0P
15	DSI_D0N	I/O	--	MIPI DSI D0N
16	PA10	I/O	Z	SD0_DATA3(B)
17	PA9	I/O	Z	SD0_DATA2(B)
18	PA8	I/O	Z	SD0_DATA1(B)
19	PA7	I/O	Z	SD0_CLK(B)
20	PA6	I/O	Z	SD0_CMD(B)
21	PA5	I/O	Z	SD0_DATA0(B)
22	WVSS	G	--	Ground of Wireless
23	WVDD14	P	--	Wireless 1.4V Power
24	WVDD33	P	--	Wireless 3.3V Power
25	ANTP	RF	--	Antenna Positive Port
26	WVDD14	P	--	Wireless 1.4V Power
27	XOSCI	I	--	Crystal Oscillator Input
28	XOSCO	O	--	Crystal Oscillator Output
29	PE13	I/O	Z	--
30	PE12	I/O	Z	--
31	PE11	I/O	Z	--
32	PE10	I/O	Z	--
33	IOVDD2	P	--	IO Power for PE10~PE13
34	PE5	I/O	Z	ADC8(ADC Input Channel 8) IO Wakeup Channel 8 Sensor0_SYNC1(A)

Pin No.	Name	Type	IO Initial State	Description
35	PE4	I/O	Z	ADC7(ADC Input Channel 7) IO Wakeup Channel 7 Sensor0_SYNC0(A)
36	PE3	I/O	Z	SensorX_CLK Sensor0_CLK(A)
37	PE2	I/O	Z	SensorX_D7 Sensor0_D9(A) Ethnet RMII_RX1(A)
38	PE1	I/O	Z	SensorX_D6 Sensor0_D8(A) Ethnet RMII_RX0(A)
39	PE0	I/O	Z	SensorX_D5 Sensor0_D7(A) Ethnet RMII_CRSDV(A) SD1_CLK(B)
40	PD15	I/O	Z	SensorX_D4 Sensor0_D6(A) Ethnet RMII_REFCLK(A) SD1_CMD(B)
41	PD14	I/O	Z	SensorX_D3 Sensor0_D5(A) Ethnet RMII_RXERR(A) SD1_DATA0(B)
42	PD13	I/O	Z	SensorX_D2 Sensor0_D4(A) Ethnet RMII_TXEN(A) SD1_DATA1(B)
43	PD12	I/O	Z	SensorX_D1 Sensor0_D3(A) Ethnet RMII_TX1(A) SD1_DATA2(B)
44	PD11	I/O	Z	SensorX_D0 Sensor0_D2(A) Ethnet RMII_TX0(A) SD1_DATA3(B)
45	PD10	I/O	Z	ADC6(ADC Input Channel 6) IO Wakeup Channel 6 Sensor0_D1(A)
46	USBAVDD	P	--	High Speed USB Power
	IOVDD	P	--	IO Power for PA5~PA10, PA12~PA15, PB0~PB7, PB10~PB15, PC0, PC1, PC6~PC13, PD1, PD8, PD10~PD15, PE0~PE5, PF0~PF5, PV0~PV1

Pin No.	Name	Type	IO Initial State	Description
47	HUSBDM	I/O	15kΩ Pull-down	High Speed USB Negative Data
48	HUSBDP	I/O	15kΩ Pull-down	High Speed USB Positive Data
49	PD8	I/O	10kΩ Pull-up	MCLR(Device Reset) ADC4(ADC Input Channel 4) SD Power IO Wakeup Channel 4
50	DVDD	P	--	Digital Logic Power
51	PD1	I/O	10kΩ Pull-up	Hold down 0 to reset ADC1(ADC Input Channel 1) IO Wakeup Channel 1
52	PF5	I/O	Z	SFCTZ_DO SPITZ_DO
53	PF4	I/O	Z	SFCTZ_CLK SPITZ_CLK
54	PF3	I/O	Z	SFCTZ_DATA3 SPITZ_DATA3
55	FSPG	I/O	Z	Flash Power Output
	PC13	I/O	Z	ADC13(ADC Input Channel 13) IO Wakeup Channel 13
56	PF2	I/O	Z	SFCTZ_DATA2 SPITZ_DATA2
57	PF1	I/O	Z	SFCTZ_DI SPITZ_DI
58	PF0	I/O	Z	SFCTZ_CS SPITZ_CS
59	PR1	I/O	Z	32k Crystal Oscillator Output
60	PR0	I/O	Z	32k Crystal Oscillator Input
61	RTCVDD33	P	--	RTC Power for PR0~PR1
62	PV1	I/O	Z	AVDD18
63	PV0	I/O	Z	AVDD28
64	IOVDD	P	--	IO Power
65	DCVDD	P	--	DCDC Power
66	VBAT	P	--	Battery Input
67	SW	P	--	Buck DCDC Switch Port
68	PGND	G	--	Ground of Buck DC-DC converter
69	PC12	I/O	Z	AIN_BN0(Audio ADC Negative Input)
70	PC11	I/O	Z	AIN_BP0(Audio ADC Positive Input)
71	PC10	I/O	10kΩ Pull-down	LVD(External Low Voltage Detection Input) MICBIASB (MIC Bias Output)
72	AUVSS	G	--	Audio Ground

Pin No.	Name	Type	IO Initial State	Description
73	VCM	P	--	Audio Reference Power
74	PC9	I/O	Z	MICBIASA (MIC Bias Output) Right Channel DAC Output
75	PC8	I/O	Z	Left Channel DAC Output
	PC7	I/O	Z	AIN_AP0(Audio ADC Positive Input)
	PC1	I/O	Z	LCD_DATA7(B) PAP_D7(B)
76	PC6	I/O	Z	AIN_AN0(Audio ADC Negative Input)
	PC0	I/O	Z	LCD_DATA6(B) PAP_D6(B)
77	PB15	I/O	Z	LCD_DATA5(B) PAP_D5(B) SD1_CLK(A)
78	PB14	I/O	Z	LCD_DATA4(B) PAP_D4(B) SD1_CMD(A)
79	PB13	I/O	Z	LCD_DATA3(B) PAP_D3(B) SD1_DATA0(A)
80	PB12	I/O	Z	LCD_DATA2(B) PAP_D2(B) SD1_DATA1(A)
81	PB11	I/O	Z	LCD_DATA1(B) PAP_D1(B) SD1_DATA2(A)
82	PB10	I/O	Z	LCD_DATA0(B) PAP_D0(B) SD1_DATA3(A)
83	PB7	I/O	Z	LCD_DATA15(A/B) Sensor1_D7(A) PAP_D15(A/B) SD0_CLK(D)
84	PB6	I/O	Z	LCD_DATA14(A/B) Sensor1_D6(A) PAP_D14(A/B) SD0_CMD(D)
85	PB5	I/O	Z	LCD_DATA13(A/B) Sensor1_D5(A) PAP_D13(A/B) SD0_DATA0(D)

Pin No.	Name	Type	IO Initial State	Description
86	PB4	I/O	Z	LCD_DATA12(A/B) Sensor1_D4(A) PAP_D12(A/B) SD0_DATA1(D)
87	PB3	I/O	Z	LCD_DATA11(A/B) Sensor1_D3(A) PAP_D11(A/B) SD0_DATA2(D)
88	PB2	I/O	Z	LCD_DATA10(A/B) Sensor1_D2(A) PAP_D10(A/B) SD0_DATA3(D)
89	PB1	I/O	Z	LCD_DATA9(A/B) Sensor1_D1(A) PAP_D9(A/B)
90	PB0	I/O	Z	LCD_DATA8(A/B) Sensor1_D0(A) PAP_D8(A/B)
91	PA15	I/O	Z	LCD_SYNC2(A/B) Sensor1_SYNC1(A)
92	PA14	I/O	Z	LCD_SYNC1(A/B) Sensor1_SYNC0(A) PAP_RD(A/B)
93	PA13	I/O	Z	LCD_DCLK(A/B) Sensor1_CLK(A)
94	PA12	I/O	Z	LCD_SYNC0(A/B) PAP_WR(A/B)
95	DVDD	P	--	Digital Logic Power
96	AVD33	P	--	Analog 3.3V Power
	IOVDD	P	--	IO Power

Note

1.IO initial state abbreviations Z--High resistance, H--High level, L--Low level, X--May be changed during power on.

2.Timer, CAN, MCPWM, QDEC, UART, LEDC, I²C, I²S and SPI functions can be remapped to any I/O (except PF/PR/PV/CSI/DSI).

Table 2-2-2 Pin Types Description

Pin Type	Description	Pin Type	Description
P	Power	I/O	Input or Output
G	Ground	I	Input
RF	RF antenna	O	Output

3 Electrical Characteristics

3.1 Absolute Maximum Ratings

Table 3-1 Absolute Maximum Ratings

Symbol	Parameter	Min	Max	Unit
T _{opt}	Operating temperature	-20	+85	°C
T _{stg}	Storage temperature	-65	+150	°C
VBAT	Supply Voltage	-0.3	5.5	V
IOVDD		-0.3	3.6	V
IOVDD2		-0.3	3.6	V
RTCVDD33		-0.3	3.6	V
DCVDD		-0.3	1.54	V
WVDD33		-0.3	3.6	V
WVDD14		-0.3	1.54	V
USBAVDD		-0.3	3.6	V
AVD33		-0.3	3.6	V
AVD12		-0.3	1.54	V
GPIO	Input voltage of GPIO	-0.3	3.6	V

Note

1. Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device.

3.2 ESD Ratings

Table 3-2 ESD Ratings

Parameter	Typ	Test pin	Reference standard
Human Body Mode	±4kV	All pins	JEDEC EIA/JESD22-A114
Machine Mode	±300V	All pins	JEDEC EIA/JESD22-A115
Charge Device Model	±1kV	All pins	ANSI/ESDA/JEDEC JS-002-2022

3.3 PMU Characteristics

Table 3-3 PMU Characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
VBAT	Power supply	--	2.7	3.7	5.5	V
Operating mode						
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
IOVDD	Voltage output	--	2.4	3.3	3.4	V
	Loading current	IOVDD=3.3V@VBAT = 3.9V	--	--	200	mA
AVDD28	Voltage output	--	2.5	2.8	3.2	V
	Loading current	AVDD28=2.8V@IOVDD = 3.3V	--	--	100	mA
AVDD18	Voltage output	--	1.5	1.8	2.2	V

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
	Loading current	AVDD18=1.8V@IOVDD = 3.3V	--	--	60	mA
DCVDD	Voltage output	--	--	1.4	--	V
	Loading current	DCVDD=1.4V@IOVDD = 3.3V, LDO mode	--	--	60	mA
		DCVDD=1.4V@VBAT = 3.7V, DCDC mode	--	--	180	mA
Low Power mode						
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
IOVDD	Loading current	IOVDD=3.0V@VBAT = 3.7V	--	--	10	mA

3.4 IO Characteristics

Table 3-4 IO Characteristics

Input Characteristics						
Symbol	Parameter	Conditions	IO	Min	Max	Unit
V _{IL}	Low-Level Input Voltage	IOVDD2 = 3.0V	PE10~PE13	-0.3	1.4	V
		IOVDD2 = 1.8V	PE10~PE13	-0.3	0.7	V
		IOVDD = 3.0V	PA5~PA10 PA12~PA15 PB0~PB7 PB10~PB15 PC0, PC1 PC6~PC13 PD1, PD8 PD10~PD15 PE0~PE5 PF0~PF5 HUSBDP HUSBDM PR0~PR1 PV0~PV1	-0.3	1.4	V
V _{IH}	High-Level Input Voltage	IOVDD2 = 3.0V	PE10~PE13	1.7	3.3	V
		IOVDD2 = 1.8V	PE10~PE13	1.1	2.0	V
		IOVDD = 3.0V	PA5~PA10 PA12~PA15 PB0~PB7 PB10~PB15 PC0, PC1 PC6~PC13 PD1, PD8 PD10~PD15 PE0~PE5 PF0~PF5 HUSBDP	1.7	3.3	V

			HUSBDM PR0~PR1 PV0~PV1			
Output Characteristics						
Symbol	Parameter	Conditions	IO	Typ	Unit	
I _{OL}	Output Current	IOVDD2 = 3.0V Voutput = 0.3V	PE10~PE13	2.5(HD=0) 8(HD=1) 18.5(HD=2) 24(HD=3)	mA	
		IOVDD2 = 1.8V Voutput = 0.2V				
		IOVDD = 3.0V Voutput = 0.3V	PA5~PA10 PA12~PA15 PB0~PB7 PB10~PB15 PC0, PC1 PC6~PC13 PD1, PD8 PD10~PD15 PE0~PE5 PF0~PF5	2.5(HD=0) 8(HD=1) 18.5(HD=2) 24(HD=3)	mA	
			PR0~PR1 PV0~PV1	2.5(HD=0) 18.5(HD=1)		
			HUSBDP HUSBDM	8	mA	
I _{OH}	Output Current	IOVDD2 = 3.0V Voutput = 2.7V	PE10~PE13	2.5(HD=0) 8(HD=1) 18.5(HD=2) 24(HD=3)	mA	
		IOVDD2 = 1.8V Voutput = 1.6V				
		IOVDD = 3.0V Voutput = 2.7V	PA5~PA10 PA12~PA15 PB0~PB7 PB10~PB15 PC0, PC1 PC6~PC13 PD1, PD8 PD10~PD15 PE0~PE5 PF0~PF5	2.5(HD=0) 8(HD=1) 18.5(HD=2) 24(HD=3)	mA	
			PR0~PR1 PV0~PV1	2.5(HD=0) 18.5(HD=1)		
			HUSBDP HUSBDM	8	mA	

Internal Resistance Characteristics					
Symbol	Parameter	Conditions	IO	Typ	Unit
R _{pu}	Pull-up Resistance	IOVDD = 3.0V IOVDD2 = 3.0V/1.8V	PA5~PA10 PA12~PA15 PB0~PB7 PB10~PB15 PC0, PC1 PC6~PC13 PD1, PD8 PD10~PD15 PE0~PE5 PE10~PE13 PF0~PF5	10k	Ω
			HUSB DP	1.5k(PU=1) 1k(PU=2/3)	Ω
R _{pd}	Pull-down Resistance	IOVDD = 3.0V IOVDD2 = 3.0V/1.8V	PA5~PA10 PA12~PA15 PB0~PB7 PB10~PB15 PC0, PC1 PC6~PC13 PD1, PD8 PD10~PD15 PE0~PE5 PE10~PE13 PF0~PF5	10k	Ω
			HUSB DP HUSB DM	15k	Ω

Note

1.Internal pull-up/pull-down resistance accuracy ±20%.

3.5 Audio DAC Characteristics

Table 3-5 Stereo DAC Characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Resolution	--	--	16	--	bits
Output Sample Rate	--	8	--	96	kHz
SNR	Differential Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	103	--	dB
	Single Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	100	--	dB
Dynamic Range	Differential Mode Fin=1kHz@-60dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	103	--	dB
	Single Mode Fin=1kHz@-60dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	100	--	dB
THD+N	Differential Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	-87	--	dB
	Single Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	-75	--	dB
Noise Floor	Differential Mode B/W=20Hz~20kHz A-Weighted load=10kΩ	--	6.4	--	uVrms
	Single Mode B/W=20Hz~20kHz A-Weighted load=10kΩ	--	5.4	--	uVrms

3.6 Audio ADC Characteristics

Table 3-6 Audio ADC Characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Resolution	--	--	16	--	bits
Input Sample Rate	--	8	--	48	kHz
SNR	Differential input Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	95	--	dB
	Single-ended input Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	92	--	dB
Dynamic Range	Differential input Mode Fin=1kHz@-60dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	95	--	dB
	Single-ended input Mode Fin=1kHz@-60dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	92	--	dB
THD+N	Differential input Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	-87	--	dB
	Single-ended input Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	-81	--	dB
Analogue Gain		-6	--	28	dB
Max Input Level	Differential input Mode ADC gain=0dB	--	0.7	--	Vrms
	Single-ended input Mode ADC gain=0dB	--	0.35	--	Vrms

3.7 BT Characteristics

3.7.1 Transmitter

Table 3-7-1 Transmitter characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Maximum RF Transmit Power	BR	--	19	--	dBm
Maximum RF Transmit Power	EDR $\pi/4$ DQPSK EDR 8DPSK	--	19	--	dBm
Relative Transmit Power	EDR $\pi/4$ DQPSK EDR 8DPSK	--	1.5	--	dB
Maximum RF Transmit Power	BLE-1Mbps	--	19	--	dBm

3.7.2 Receiver

Table 3-7-2 Receiver characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Sensitivity	BR	--	-95.5	--	dBm
	EDR $\pi/4$ DQPSK	--	-96	--	dBm
	EDR 8DPSK	--	-88	--	dBm
	BLE-1Mbps	--	-98	--	dBm
	BLE-2Mbps	--	-95	--	dBm
	BLE-S2	--	-101	--	dBm
	BLE-S8	--	-106	--	dBm

3.8 WiFi Characteristics

3.8.1 Transmitter

TX Power with Spectral Mask and EVM Meeting 802.11 Standards.

Table 3-8-1 Transmitter characteristics

Parameter	Conditions	Min	Typ	Max	Unit
TX Power	802.11b, 1 Mbps, DSSS	--	19	--	dBm
	802.11b, 11 Mbps, CCK	--	19	--	dBm
	802.11g, 6 Mbps, OFDM	--	17	--	dBm
	802.11g, 54 Mbps, OFDM	--	14	--	dBm
	802.11n, HT20, MCS0	--	17	--	dBm
	802.11n, HT20, MCS7	--	13	--	dBm
	802.11n, HT40, MCS0	--	17	--	dBm
	802.11n, HT40, MCS7	--	12	--	dBm

3.8.2 Receiver

For RX tests, the PER (packet error rate) limit is 8% for 802.11b, and 10% for 802.11g/n.

Table 3-8-2 Receiver characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Sensitivity	802.11b, 1 Mbps, DSSS	--	-97	--	dBm
	802.11b, 2 Mbps, DSSS	--	-94.5	--	dBm
	802.11b, 5.5 Mbps, CCK	--	-93	--	dBm
	802.11b, 11 Mbps, CCK	--	-90	--	dBm
	802.11g, 6 Mbps, OFDM	--	-93	--	dBm
	802.11g, 9 Mbps, OFDM	--	-92	--	dBm
	802.11g, 12 Mbps, OFDM	--	-91	--	dBm
	802.11g, 18 Mbps, OFDM	--	-89	--	dBm
	802.11g, 24 Mbps, OFDM	--	-86	--	dBm
	802.11g, 36 Mbps, OFDM	--	-83	--	dBm
	802.11g, 48 Mbps, OFDM	--	-79	--	dBm
	802.11g, 54 Mbps, OFDM	--	-77	--	dBm
	802.11n, HT20, MCS0	--	-93	--	dBm
	802.11n, HT20, MCS1	--	-90.5	--	dBm
	802.11n, HT20, MCS2	--	-88	--	dBm
	802.11n, HT20, MCS3	--	-84.5	--	dBm
	802.11n, HT20, MCS4	--	-81.5	--	dBm
	802.11n, HT20, MCS5	--	-77	--	dBm
	802.11n, HT20, MCS6	--	-75	--	dBm
	802.11n, HT20, MCS7	--	-74	--	dBm
	802.11n, HT40, MCS0	--	-89	--	dBm
	802.11n, HT40, MCS1	--	-86	--	dBm
	802.11n, HT40, MCS2	--	-84	--	dBm
	802.11n, HT40, MCS3	--	-80	--	dBm
	802.11n, HT40, MCS4	--	-77.5	--	dBm
	802.11n, HT40, MCS5	--	-72.5	--	dBm
	802.11n, HT40, MCS6	--	-71.5	--	dBm
	802.11n, HT40, MCS7	--	-70	--	dBm
	802.11n, HT40, MCS32	--	-89	--	dBm

4 Package Information

4.1 QFN96_10*10mm

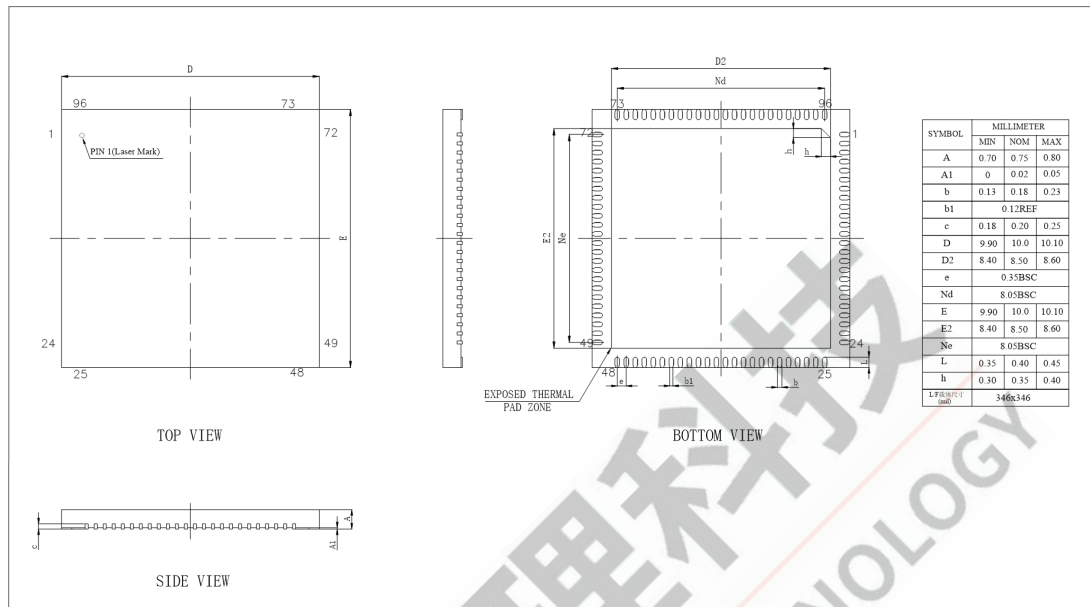


Figure 4-1 AC7922A Package

5 IC Marking Information

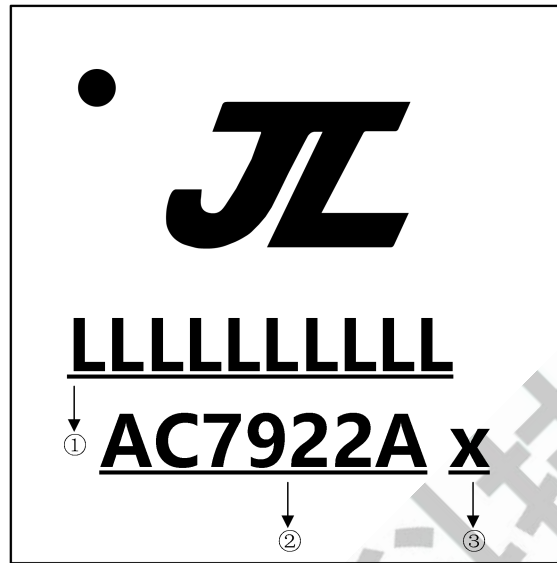


Figure 5-1 AC7922A Package Outline

- ① Production Batch
- ② Chip Model
- ③ Built-in DDR size
 - 0 No Flash Memory
 - 2 2Mbit flash
 - 4 4Mbit flash
 - 8 8Mbit flash
 - 6 16Mbit flash
 - 3 32Mbit flash
 - 5 64Mbit flash
 - 7 128Mbit flash
 - A 1Mx16 SDRAM
 - B 4Mx16 SDRAM
 - E 4Mx16bit DDR1
 - F 8Mx16bit DDR1
 - G 16Mx16bit DDR1

6 Solder-Reflow Condition

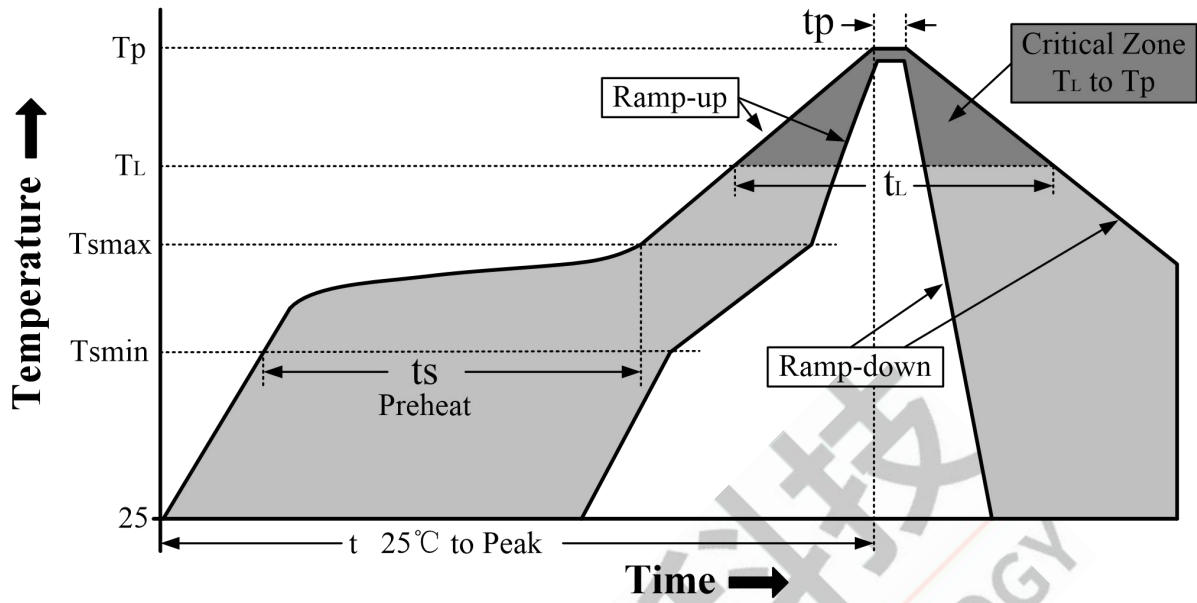


Figure 6-1 Classification Reflow Profile

Table 6-1 Classification Profiles

Profile Feature		Sn-Pb Eutectic Assembly	Pb-Free Assembly
Preheat/Soak	Temperature Min (T_{smin})	100°C	150°C
	Temperature Max (T_{smax})	150°C	200°C
	Time (t_s) from (T_{smin} to T_{smax})	60-120 seconds	60-180 seconds
Average ramp-up rate (T_{smax} to T_p)		3°C/second max	3°C/second max
Liquidous temperature (T_L)		183°C	217°C
Time (t_L) maintained above T_L		60-150 seconds	60-150 seconds
Peak package body temperature (T_p)		See Table 6-2	See Table 6-3
Time within 5°C of actual Peak Temperature (t_p) ²		10-30 seconds	20-40 seconds
Ramp-down rate (T_p to T_L)		6°C/second max	6°C/second max
Time 25°C to peak temperature		6 minutes max	8 minutes max

Note

- 1.All temperatures refer to topside of the package, measured on the package body surface
- 2.Time within 5°C of actual peak temperature (t_p) specified for the reflow profiles is a "supplier" and "user" maximum.

Table 6-2 SnPb Classification Temperature

Package Thickness	Volume mm ³ < 350	Volume mm ³ ≥ 350
<2.5 mm	240 +0/-5°C	225 +0/-5°C
≥2.5 mm	225 +0/-5°C	225 +0/-5°C

Table 6-3 Pb-free - Classification Temperature

Package Thickness	Volume mm ³ < 350	Volume mm ³ 350 - 2000	Volume mm ³ > 2000
< 1.6mm	260°C	260°C	260°C
1.6 mm - 2.5mm	260°C	250°C	245°C
> 2.5mm	250°C	245°C	245°C

Note

1.*Tolerance The device manufacturer/supplier shall assure process compatibility up to and including the stated classification temperature (this means Peak reflow temperature +0°C.For example 260°C+0°C)at the rated MSL level.